Supplementary Material for "FundusShift: Benchmarking Accuracy and Reliability of Fundus Disease Classification at Unseen Sources, and a Low-Capacity Remedy"

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S1 Data, Leakage Audit and Ethics

Of 40,414 inventoried images, 35,280 enter the protocol, and 5,134 are unused for reasons recorded in the dataset files. The audit of the released splits found zero violations for: leakage group across splits; identical file (SHA-256) across splits; patient across splits; near-duplicate pair across splits; acquisition family on both development and test sides; and source on both sides. Brightness-altered copies of GRAPE photographs in the FIVES glaucoma class were removed.

Label scope. Grading scales differ between sources, and N means "no DR" in DR-only collections, so a normal label does not exclude other disease there. Patient identifiers exist only for some collections. Within several collections, trivial image statistics predict the class (for example FIVES); quality matching reduced this only in part. A pilot with AI raters was too weak on glaucoma to estimate label error, and no label was changed. No independent clinical re-grading was done.

Excluded sources. RFMiD 2.0 was used for inventory and duplicate checks only, as class is confounded with camera. A glaucoma collection of optic-disc crops was excluded for a different field of view.

Ethics and licences. Licences follow each source, and the full benchmark is non-commercial because it includes OIA-DDR (CC BY-NC-SA 4.0). Published ethics and consent statements were found for FIVES, GRAPE, PAPILA, SUSTech-SYSU, DeepDRiD, Jichi, HYGD, Chákṣu, PALM and Paraguay, to the extent stated by each. Statements were not found on the release pages of HRF, Bangladesh Eye and AFIO, and the OIA-DDR and IDRiD papers were not read in full for this purpose.

S2 Development Search and Candidates

All values in this section are leave-one-source-out on development data with equal-site means, using no test row. Three seeds unless noted. Table S1 specifies LANTERN. Table S2 lists designs that were tried and not retained. Table S3 lists the ten configurations with development bAcc of at least 0.791, all evaluated on test. LANTERN was adopted for its test bAcc, and development rank did not predict test rank.

Table S1. Complete specification of LANTERN.

Encoder	MedSigLIP (patch 14, 448 px, 27 layers, width 1152), frozen
Head input	Last hidden state, 32×32 tokens pooled to 14×14; fixed random orthonormal map to 384 dimensions (seed 0); grid mean
Standardization	Per dimension, training-split mean and standard deviation
Projection	Within-class between-site means, at most 32 directions, fitted per head on its training rows
Heads	One linear head per development source, trained without it; 1,155 parameters each
Training	Label smoothing 0.1; class-balanced sampling; AdamW, lr 3×10^{-4}, wd 10^{-2}; batch 64; 12 epochs; best epoch by validation bAcc on the head's own sites
Rescaling	Centre per image; divide by the standard deviation over validation rows
Text	Mean of clip_ensemble, cupl and descriptive_single banks, on the pooled image embedding
Fusion	0.5 heads + 0.5 text
Post-processing	TS or FTS on Cal-A; thresholds and conformal quantiles on Cal-B
Seeds	0, 1, 2 on the published split; one per rotation

Table S2. Designs tried and not retained. Reference: pooled-feature bag, projection and text 0.791; single cross-entropy head on pooled features 0.675.

Experiment	bAcc	Reading
Batch network predicting temperature and class bias (1 seed)	.539	Below the raw head (.666)
EM prior shift with TS (1 seed)	.523	Unreliable with few classes per source
Same-site conditional-logit pair loss, single head	.723	Helps alone (.675); no gain in full pipeline (.792)
Polar Fourier features around the disc	.702	Below pair loss alone (.723)
Site prior, κ from 30 to 3000	.693–.717	No gain
Full residual corrector over logits	.667	Recall G fell to 0
Corrector with fusion weights only	.781	ECE .057, bAcc did not rise
RPAttn head on three token depths (1 seed)	.664	Recall G .008; with text .690
RPAttn + MixStyle (1 seed)	.614	Recall G .012; with text .688
N/D from RPAttn, G from low-capacity branch (1 seed)	.788	.793 for the pooled pipeline on that seed
Means of three token depths in full pipeline	.731	Shallow layers increased false G
Late fusion of nine encoder inputs	.765	Other encoders lowered the score
Greedy nested encoder selection	.789	Mostly kept MedSigLIP alone
Magnified optic-disc view with whole image	.768	Recall G up (.767), false G up (.135)
Image-conditioned fusion gate	.763	Below a plain average
Global class offset chosen on other sites	.679	GRAPE fell to .006
ConvNeXt-T end-to-end with pair loss (1 seed)	.622	GRAPE .004; plain fine-tuning arm stopped early

S3 Calibration States of All Representatives

Table S4 gives raw, TS and FTS on one row, on the published split. FTS seldom improves on TS. For the MedSigLIP linear probe, TS increases ECE. The best frozen probe by test score, a class-weighted cross-entropy MedSigLIP probe, had test bAcc 0.778, equal-source 0.702, raw ECE 0.065 and recall G 0.657. It was not a validation choice and is listed for transparency.

S4 Source Rotations and Pairwise Discrimination

Each rotation draws 7 of the 14 sources as test, requiring all three classes on each side. Development rows are re-split 80/10/10 by leakage group, and Cal is split into two halves. Internal estimates use a held-back half of Cal-B. LANTERN uses the frozen recipe, one seed per rotation; on the published partition this code reproduced seed 0 (0.814). Across the 50 rotations and over the eight probes, validation exceeded test in

Table S3. Candidates with development bAcc of at least 0.791. Dev: leave-one-source-out. Eq-src: equal-source mean. NLL after TS. Mirror: trained and averaged over an image and its mirror.

Configuration	Dev	Test bAcc	Eq-src	Recall G	NLL	Chákṣu
Mirror, weight decay 10^{-3}	.800	.809	.726	.696	.503	.714
Mirror, weight decay 10^{-2}	.799	.808	.727	.708	.502	.726
Mirror, weight decay 10^{-4}	.795	.807	.724	.681	.511	.703
Cross-site pair loss	.794	.809	.717	.722	.507	.694
Same-site pair loss	.792	.791	.696	.670	.524	.713
Tokens, weight decay 10^{-2} (LANTERN)	.792	.817	.724	.765	.482	.731
Pooled feature, weight decay 10^{-4}	.791	.799	.706	.687	.520	.684
Pooled feature + tokens	.791	.812	.719	.719	.493	.701
Disc side canonicalised	.791	.805	.733	.679	.511	.637
Mirrored text only	.791	.800	.714	.679	.512	.704

98% of rows, risk exceeded internal risk in 89%, and LAC fell below internal coverage in 84%. For LANTERN these were 94%, 88% and 94%. Its worst source on each rotation averaged 0.575.

S5 Attempted Fixes, Error Analysis and Disc-Side Bias

Target labels. With K labels from the target source, averaged over sources that have enough images, re-fitting the conformal threshold brought LAC coverage to about 0.90 for K from 10 to 100 (fine-tune: from 0.59). Variability is large: the worst 5% of draws covered about 0.71 at $K = 10$ and 0.84 at $K = 100$. These techniques change reliability, and change bAcc little.

Error analysis. Table S9 covers the pooled-feature pipeline on held-out development sites. On OIA-DDR, 1,629 of 5,639 normal images were predicted as G, although the source holds no glaucoma. A global class offset chosen on other sites lowered mean bAcc to 0.679, so a threshold does not fix the ranking of the G score. Between 0.9% and 13.7% of a site's images were wrong under all nine encoder inputs, and among normal IDRiD images, 28%.

Disc-side bias. Table S10 uses the same pipeline, with the disc located by an automatic, label-free rule. Images with the disc near the centre are omitted. Training on images and mirrors, on development data, lowered false G from 0.070 to 0.048, moved the normal-as-G rates to 0.078 (left) and 0.077 (right), and lowered recall G from 0.639 to 0.538. Heavier regularization partly restored recall (0.622 to 0.701). Mirrored variants appear in Table S3.

Later attempts. Two further components were tried with selection rules written before their results were read (`docs/protocol.md`). A label-scope loss treating normal labels from DR-only sources as {N, G} failed its rule on the rotations (main text, Sec. 7). A leave-one-source-out reference for thresholds kept coverage and risk promises

Table S4. Raw/TS/FTS on test, and ECE on Chákṣu.

Method	ECE	NLL	Brier	Chákṣu ECE
LANTERN	.176/.020/.017	.591/.482/.484	.323/.269/.269	.151/.035/.042
16-shot probe, MedSigLIP	.096/.044/.037	.701/.644/.644	.389/.371/.371	.096/.060/.061
Linear probe, MedSigLIP	.024/.079/.093	.647/.694/.739	.357/.367/.373	.081/.140/.159
Zero-shot, MedSigLIP	.207/.032/.032	.838/.690/.690	.488/.408/.408	.053/.214/.214
ERM, DINOv2 LoRA	.154/.115/.119	1.118/.944/.965	.479/.462/.464	.165/.113/.117
Group DRO, ConvNeXt-T	.082/.091/.101	.886/.926/1.042	.413/.416/.420	.073/.081/.086
Fine-tune (NLL rule), ConvNeXt-T	.185/.133/.147	1.347/1.036/1.158	.504/.478/.487	.271/.211/.225
CORAL, ConvNeXt-T	.198/.136/.148	1.378/1.005/1.113	.518/.484/.493	.274/.206/.218
SWAD, ConvNeXt-T	.208/.139/.149	1.566/1.056/1.164	.522/.483/.491	.301/.219/.230
Fine-tune, Swin-T	.223/.171/.182	1.563/1.174/1.311	.566/.534/.543	.306/.239/.257
WiSE-FT, SigLIP2-L	.236/.168/.179	1.842/1.243/1.382	.598/.557/.565	.168/.084/.101
MLP probe, DINOv2-B	.263/.172/.183	2.308/1.259/1.421	.625/.554/.563	.324/.215/.227
CoOp ctx-init, BiomedCLIP	.287/.150/.160	1.764/1.189/1.220	.745/.653/.657	.223/.170/.168
CoOp + LP, SigLIP-B	.243/.176/.179	1.368/1.180/1.166	.715/.673/.671	.271/.203/.217
Tip-Adapter-F, BiomedCLIP	.225/.147/.152	1.284/1.099/1.110	.733/.672/.677	.140/.165/.166
CoOp, SigLIP-B	.236/.175/.177	1.346/1.180/1.163	.712/.676/.673	.251/.199/.205
Tip-Adapter, BiomedCLIP	.332/.091/.096	1.747/1.020/1.031	.841/.628/.636	.245/.170/.164

Table S5. Fifty rotations, mean per method. Last column: rotations with test risk above internal risk.

Method	Val	Test	ECE raw	ECE TS	Risk int.	Risk test	LAC int.	LAC test	Risk up
LANTERN	.853	.771	.146	.033	.080	.116	.898	.853	88%
MedSigLIP full	.922	.745	.095	.084	.042	.134	.901	.769	100%
MedSigLIP 16-shot	.814	.716	.123	.056	.164	.217	.899	.868	74%
DINOv2-B full	.892	.655	.123	.118	.077	.221	.899	.735	100%
DINOv2-B 16-shot	.713	.552	.192	.073	.296	.383	.900	.860	78%
SigLIP2-L full	.876	.631	.155	.141	.102	.280	.900	.728	100%
SigLIP2-L 16-shot	.722	.578	.160	.080	.292	.372	.901	.871	74%
ConvNeXt-T full	.860	.593	.149	.130	.112	.287	.901	.730	100%
ConvNeXt-T 16-shot	.679	.494	.223	.091	.352	.463	.900	.840	86%

far more often but failed both pre-written efficiency rules (main text, Sec. 6.1). Both are reported in the main text.

S6 Paired Comparisons and Conformal Sets

Differences are LANTERN minus baseline. For ECE and NLL a negative difference favours LANTERN. Holm correction used 18 comparisons, including two accuracy-rule Tip-Adapter rows identical to those shown; the smallest attainable adjusted p with 2,000 draws is 0.009. Baselines with duplicates are listed once.

Table S11: Paired bootstrap over leakage groups, published split.

Metric	Baseline	Baseline value	Diff.	95% CI	Adj. p
bAcc (raw)	16-shot probe, MedSigLIP	.748	+0.069	[+0.054, +0.084]	0.009

Table S11: Paired bootstrap over leakage groups, published split.

Metric	Baseline	Baseline value	Diff.	95% CI	Adj. p
bAcc (raw)	Linear probe, MedSigLIP	.728	+0.089	[+0.073, +0.103]	0.009
bAcc (raw)	Zero-shot, MedSigLIP	.709	+0.108	[+0.093, +0.124]	0.009
bAcc (raw)	ERM, DINOv2 LoRA	.707	+0.109	[+0.093, +0.125]	0.009
bAcc (raw)	Group DRO, ConvNeXt-T	.707	+0.110	[+0.096, +0.125]	0.009
bAcc (raw)	Fine-tune (NLL rule), ConvNeXt-T	.677	+0.140	[+0.123, +0.156]	0.009
bAcc (raw)	CORAL, ConvNeXt-T	.676	+0.140	[+0.123, +0.157]	0.009
bAcc (raw)	SWAD, ConvNeXt-T	.672	+0.145	[+0.127, +0.161]	0.009
bAcc (raw)	Fine-tune, Swin-T	.655	+0.162	[+0.143, +0.179]	0.009
bAcc (raw)	WiSE-FT, SigLIP2-L	.634	+0.182	[+0.164, +0.200]	0.009
bAcc (raw)	MLP probe, DINOv2-B	.598	+0.218	[+0.198, +0.237]	0.009
bAcc (raw)	CoOp ctx-init, BiomedCLIP	.497	+0.320	[+0.301, +0.338]	0.009
bAcc (raw)	CoOp + LP, SigLIP-B	.496	+0.321	[+0.303, +0.339]	0.009
bAcc (raw)	Tip-Adapter-F, BiomedCLIP	.495	+0.322	[+0.301, +0.343]	0.009
bAcc (raw)	CoOp, SigLIP-B	.491	+0.325	[+0.307, +0.344]	0.009
bAcc (raw)	Tip-Adapter, BiomedCLIP	.478	+0.339	[+0.318, +0.359]	0.009
ECE (TS)	Zero-shot, MedSigLIP	.032	-0.012	[-0.025, +0.005]	0.221
ECE (TS)	16-shot probe, MedSigLIP	.044	-0.024	[-0.032, -0.009]	0.009
ECE (TS)	Linear probe, MedSigLIP	.079	-0.060	[-0.071, -0.038]	0.009
ECE (TS)	Group DRO, ConvNeXt-T	.091	-0.072	[-0.084, -0.052]	0.009
ECE (TS)	Tip-Adapter, BiomedCLIP	.091	-0.072	[-0.083, -0.052]	0.009
ECE (TS)	ERM, DINOv2 LoRA	.115	-0.096	[-0.108, -0.075]	0.009
ECE (TS)	Fine-tune (NLL rule), ConvNeXt-T	.133	-0.114	[-0.126, -0.092]	0.009
ECE (TS)	CORAL, ConvNeXt-T	.136	-0.116	[-0.130, -0.094]	0.009
ECE (TS)	SWAD, ConvNeXt-T	.139	-0.119	[-0.133, -0.096]	0.009
ECE (TS)	Tip-Adapter-F, BiomedCLIP	.147	-0.127	[-0.142, -0.108]	0.009
ECE (TS)	CoOp ctx-init, BiomedCLIP	.150	-0.131	[-0.145, -0.107]	0.009
ECE (TS)	WiSE-FT, SigLIP2-L	.168	-0.149	[-0.163, -0.126]	0.009
ECE (TS)	Fine-tune, Swin-T	.171	-0.151	[-0.165, -0.128]	0.009
ECE (TS)	MLP probe, DINOv2-B	.172	-0.152	[-0.166, -0.127]	0.009
ECE (TS)	CoOp, SigLIP-B	.175	-0.156	[-0.171, -0.131]	0.009
ECE (TS)	CoOp + LP, SigLIP-B	.176	-0.156	[-0.172, -0.132]	0.009
NLL (TS)	16-shot probe, MedSigLIP	.644	-0.161	[-0.186, -0.136]	0.009
NLL (TS)	Zero-shot, MedSigLIP	.690	-0.208	[-0.227, -0.188]	0.009
NLL (TS)	Linear probe, MedSigLIP	.694	-0.212	[-0.248, -0.173]	0.009
NLL (TS)	Group DRO, ConvNeXt-T	.926	-0.444	[-0.511, -0.377]	0.009
NLL (TS)	ERM, DINOv2 LoRA	.944	-0.462	[-0.526, -0.404]	0.009
NLL (TS)	CORAL, ConvNeXt-T	1.005	-0.523	[-0.589, -0.460]	0.009
NLL (TS)	Tip-Adapter, BiomedCLIP	1.020	-0.537	[-0.567, -0.506]	0.009
NLL (TS)	Fine-tune (NLL rule), ConvNeXt-T	1.036	-0.553	[-0.624, -0.485]	0.009
NLL (TS)	SWAD, ConvNeXt-T	1.056	-0.574	[-0.647, -0.502]	0.009
NLL (TS)	Tip-Adapter-F, BiomedCLIP	1.099	-0.617	[-0.652, -0.581]	0.009
NLL (TS)	Fine-tune, Swin-T	1.174	-0.691	[-0.766, -0.619]	0.009
NLL (TS)	CoOp + LP, SigLIP-B	1.180	-0.698	[-0.752, -0.644]	0.009
NLL (TS)	CoOp, SigLIP-B	1.180	-0.698	[-0.751, -0.645]	0.009
NLL (TS)	CoOp ctx-init, BiomedCLIP	1.189	-0.707	[-0.765, -0.649]	0.009
NLL (TS)	WiSE-FT, SigLIP2-L	1.243	-0.761	[-0.837, -0.684]	0.009
NLL (TS)	MLP probe, DINOv2-B	1.259	-0.777	[-0.873, -0.681]	0.009

S7 Recipes, Multimodal LLMs, Open Set and Reproducibility

Recipe audit. Fine-tuning recipes were audited before final runs: ResNet-50 runs at 224 px rather than 176 px, Swin-T uses no drop-path, and images are resized square before cropping in training. Earlier runs were archived and not used. SWAD's rows in the feature-based out-of-distribution analysis were invalid, since its saved checkpoint held pre-averaging weights, and are not reported.

Table S6. Pairwise AUROC, mean over sources holding both classes, published split. Comparisons rest on 3 or 4 sources and are not corrected for multiplicity.

Pair (sources)	16-shot	Linear probe	Group DRO	ERM	Fine-tune
D–G (3)	.939	.917	.942	.916	.916
N–D (4)	.946	.928	.923	.909	.900
N–G (4)	.749	.678	.704	.644	.643

Table S7. Test-time adaptation, change in test bAcc. A dash marks a method that does not apply.

Model	Baseline	TENT	BN	T3A K=20	T3A K=100
Fine-tune Swin-T	.655	+.000	–	+.011	+.013
Fine-tune ResNet-50	.661	−.063	−.063	−.030	−.020
Group DRO ConvNeXt-T	.707	+.003	–	−.002	−.004
ERM DINOv2 LoRA	.708	+.003	–	−.064	−.046
16-shot MedSigLIP	.748	–	–	−.089	−.065
Linear probe MedSigLIP	.728	–	–	−.049	−.049

Open set. 861 images outside N, D and G form a probe (AFIO 24, FIVES 200, PALM 637). At an 80% Cal-B coverage threshold, maximum softmax accepted 34% to 68% of them across the probes and models examined. A KNN feature score cut this to 6% to 22%, at test coverage of 0.40 to 0.55. This lower risk largely comes from dropping hard sources altogether, so it is useful for flagging strange images rather than as a remedy for shift. LANTERN was not evaluated on the probe.

Preprocessing and other checks. CLAHE did not help fine-tuning (ConvNeXt-T 0.668 to 0.653, Swin-T 0.655 to 0.650). The MedSigLIP linear probe rose from 0.728 to 0.788 on test, while its validation fell (0.920 to 0.911), so selection by validation would not have chosen it.

Reproducibility. Tables and figures are generated from saved results by `paper/make_assets.py`. Method code is in `experiments/top_dev_final.py`, rotations in `experiments/lantern_rotation.py`, paired tests in `experiments/proposed_paired.py`, and the protocol with recorded deviations in `docs/protocol.md`. URLs: [public URL, DOI, version].

References

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3. Li, C., Wong, C., Zhang, S., et al.: LLaVA-Med: Training a large language-and-vision assistant for biomedicine in one day. In: Advances in Neural Information Processing Systems, vol. 36 (2023)
4. Bai, S., et al.: Qwen2.5-VL technical report. arXiv:2502.13923 (2025)

Table S8. Prior correction, test bAcc. Oracle uses each source's true class rates and is not a method; single-class HYGD and PALM make the gain easier.

Method	TS	EM per source	True pooled prior	Oracle per source
16-shot	.748	.787	.769	.853
Linear probe	.728	.835	.778	.853
Zero-shot	.709	.635	.709	.863
Group DRO	.707	.723	.733	.857
ERM	.707	.726	.727	.826
Fine-tune	.655	.661	.675	.827
WiSE-FT	.634	.627	.669	.813

Table S9. Held-out development sites: bAcc of head, text and fusion, recalls of fusion, and images wrong under all nine encoder inputs.

Held-out site	*n*	Head	Text	Fusion	N	D	G	All wrong
Bangladesh Eye	3,153	.651	.732	.753	.761	.963	.536	.039
OIA-DDR	11,269	.660	.738	.735	.685	.786	–	.137
DeepDRiD	1,804	.803	.721	.795	.682	.908	–	.111
GRAPE	528	.644	.750	.741	–	–	.741	.104
IDRiD	462	.757	.752	.751	.527	.974	–	.100
Jichi	8,475	.779	.736	.803	.824	.781	–	.103
SUSTech-SYSU	1,097	.975	.889	.957	.938	.975	–	.009

Table S10. Normal images predicted as G, by disc side, held-out site.

Held-out site	Disc left	Disc right
OIA-DDR	.386	.191
Bangladesh Eye	.141	.023
IDRiD	.119	.045
SUSTech-SYSU	.086	.019
Jichi	.067	.054
DeepDRiD	.096	.086

Table S12. LANTERN conformal sets after TS, quantiles from Cal-B.

View	Nominal	Method	Coverage	Mean size	Singleton share
Test	0.90	LAC	.885	1.20	.807
Test	0.90	ClassLAC	.884	1.25	.762
Test	0.90	APS	.955	1.89	.115
Test	0.95	LAC	.958	1.60	.516
Test	0.95	ClassLAC	.944	1.64	.514
Test	0.95	APS	.958	1.97	.032
Chákṣu	0.90	LAC	.828	1.43	.576
Chákṣu	0.90	ClassLAC	.789	1.53	.508
Chákṣu	0.90	APS	.952	2.00	.001
Chákṣu	0.95	LAC	.977	2.17	.096
Chákṣu	0.95	ClassLAC	.938	2.35	.113
Chákṣu	0.95	APS	.952	2.00	.000

Table S13. Multiple choice with multimodal LLMs, mean over six answer orders. These give discrete answers, so no TS or FTS applies. MedGemma was not accessible.

Model	Test bAcc	Recall N	Recall D	Recall G	Chákṣu bAcc
Fundus-R1 [1]	.576	.789	.795	.143	.463
Qwen2.5-VL [4]	.424	.926	.347	.000	.488
LLaVA-OneVision [2]	.407	.367	.855	.000	.096
LLaVA-Med [3]	.347	.905	.056	.080	.507